Semantic Emergence in Language Model Embeddings

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Abstract

This study examines whether the ability of language model embeddings to encode semantic structure emerges discontinuously with increasing model scale. Using a corpus of 500 books spanning 20 genres from Project Gutenberg, we analyze chunk-level embeddings produced by four model families — Pythia (70M–12B), Cerebras-GPT (111M–13B), Qwen2.5 (0.5B–7B), and OpenAI `text-embedding-3` — across 22 model checkpoints, complemented by Qwen3-Embedding as a contrastive retrieval reference. We conduct two complementary label-free analyses at the corpus level: a within-vs-across books cosine similarity pipeline that tests whether same-book passage embeddings are geometrically closer to each other than to random other-book passages, and a BERTopic topic discovery pipeline that evaluates whether embedding-driven topic structure becomes more coherent with scale. We then drill into a single genre (Romance, 25 books, ∼3,000 chunks per checkpoint) along five independent angles. The within-vs-across analysis reveals a sharp emergence threshold in Pythia between 160M and 410M parameters, where the within-vs-across gap increases approximately 28-fold in a single scale step. This threshold is independently corroborated by the BERTopic noise fraction, the within-vs-genre gap, the nearest-neighbour subject lift, and a discontinuous drop in mean top-1 cosine similarity signalling the resolution of anisotropic embedding collapse. The Romance deep-dive further shows that Pythia emergence proceeds in two sequential phases: a geometric phase at 160M→410M in which anisotropy resolves, followed by a categorical phase at 410M→1B in which discovered topic structure aligns with bookshelf and book labels. Cerebras-GPT exhibits discriminative embeddings at its smallest measured size (111M), indicating an architecture-dependent threshold below 111M, while Qwen2.5 exhibits a qualitatively different embedding geometry with uniformly compressed cosine similarities. These results demonstrate that representational emergence is quantitatively locatable, method-robust, and architecture-dependent, with implications for how model scale and training regime shape the internal organization of text.

Keywords: Text Embeddings, Semantic Emergence, Cosine Similarity, BERTopic, Topic Modeling, Language Model Scaling, Anisotropy, Nearest-Neighbour Retrieval

1 Introduction

A central question in the study of large language models is whether increased scale gives rise to qualitatively new capabilities, or whether improvements are strictly gradual and continuous. The concept of emergence — where a capability appears discontinuously as a function of scale — has been documented in task performance benchmarks [14], but remains less well characterized at the level of internal representations. Embedding spaces encode the geometric structure through which a model organizes text, and shifts in this geometry may precede or underlie observable task-level changes.

This study focuses on a specific and measurable form of representational structure: whether a model's chunk-level embeddings encode the identity of the source book at the passage level, and whether they encode the semantic structure of a single narrative genre. The first question is operationalized through a within-vs-across books comparison that tests whether embeddings from same-book chunks are more similar to each other than to embeddings from other books — a direct probe of whether book identity is encoded in the embedding geometry. The second question is operationalized through a Romance genre deep-dive that combines BERTopic topic discovery, nearest-neighbour analysis on bookshelf and Library of Congress subject labels, within-vs-across genre compactness, and LLM-based topic content labelling.

All analyses are entirely label-free in the sense that the embedding signal is never given access to genre or subject metadata — labels are used only *post hoc* to score whether the unsupervised embedding geometry recovers them. This makes the approach applicable across corpora and model families without requiring curated supervision.

Our key contributions are:

- A within-vs-across books analysis pipeline that quantifies the book-identity signal in chunk embeddings across 22 model checkpoints from four families, identifying a sharp emergence threshold at approximately 400M parameters in Pythia.
- A BERTopic topic discovery pipeline operating directly on pre-computed chunk embeddings rather than raw text, which makes it sensitive to the model's semantic organization rather than surface word statistics.
- A Romance genre deep-dive along five independent angles — BERTopic, nearest-neighbour lift, cross-corpus retrieval, within-vs-across genre compactness, and LLM-labelled topic content — that converges on the same scale boundary identified at the corpus level.
- Evidence that Pythia emergence proceeds in two sequential phases: a *geometric* phase between 160M and 410M (anisotropy resolves) and a *categorical* phase between 410M and 1B (discovered topics align with bookshelf and book labels).
- A demonstration that two independent corpus-level methods — cosine gap analysis and BERTopic noise fraction — and four additional Romance-level signals all flag the same Pythia transition zone, providing method-robust evidence for a genuine representational transition.
- Comparative analysis across Pythia, Cerebras-GPT, Qwen2.5, OpenAI `text-embedding-3`, and Qwen3-Embedding showing that the location of the emergence threshold is architecture-dependent rather than a universal function of parameter count, and that embedding-trained models occupy qualitatively different regimes from base language models.

The remainder of this report is organized as follows. Section 2 reviews related work. Section 3 describes the corpus and model families. Section 4 details the methodologies for each analysis. Section 5 presents corpus-level results, and Section 6 presents the Romance genre deep-dive. Section 7 discusses findings and their implications. Section 8 outlines future research directions, and Section 9 concludes.

2 Related Work

2.1 Emergent Capabilities in Language Models

Wei et al. [14] demonstrated that certain task capabilities appear sharply at specific model scales, motivating the treatment of emergence as a qualitative rather than purely quantitative shift. Subsequent work has debated whether emergence is a genuine discontinuity or an artifact of evaluation metrics [13]. This study contributes an embedding-level perspective that measures continuous geometric quantities (cosine similarity) rather than binary task outcomes, making it less susceptible to metric-induced discontinuities. The threshold we observe is sharp in magnitude, not in the binary sense.

2.2 Embedding Geometry and Anisotropy

Ethayarajh [5] showed that contextual representations from large language models are anisotropic — vectors tend to occupy a narrow cone in embedding space rather than spreading uniformly, causing cosine similarities between random pairs to be unexpectedly high. Cai et al. [3] connected anisotropy to the degeneration of sentence embeddings, an effect first characterized as the representation degeneration problem by Gao et al. [6], and Mu & Viswanath [10] showed that simply removing the dominant directions substantially improves embedding isotropy. The near-unity cosine similarities observed at small Pythia sizes (70M: within = 0.996, across = 0.996) are consistent with severe anisotropy: all chunk vectors collapse toward the same direction, rendering cosine similarity uninformative. The threshold at 410M corresponds to the resolution of this collapse and is independently visible as an 11-fold drop in mean top-1 cosine similarity (Section 6.3).

2.3 Semantic Similarity and Retrieval

The use of cosine similarity between contextualized embeddings as a measure of semantic relatedness is well established [12]. BERTScore [15] extends this to document-level evaluation by matching words in candidate and reference texts through cosine similarity using pre-trained BERT embeddings. Our within-vs-across analysis applies a related idea at the book level, asking whether same-book passages are more semantically coherent with each other than with random other-book passages. The nearest-neighbour lift analysis in the Romance deep-dive (Section 6.2) is in the same lineage and uses Library of Congress subject metadata as a richer label axis.

2.4 Neural Topic Modeling with BERTopic

BERTopic [7] combines neural embedding representations with density-based clustering (HDBSCAN [8]) and class-based TF-IDF (c-TF-IDF) to produce interpretable topic labels. Unlike LDA [2], which models documents as bags of words and requires specifying topic count *a priori*, BERTopic leverages semantic similarity to discover natural cluster structures of varying granularity. The noise fraction produced by HDBSCAN — the proportion of documents assigned to the outlier class (Topic -1) — serves in our analysis as a measure of how well-structured the input embedding space is. UMAP [9] is used as the dimensionality-reduction stage prior to clustering.

3 Dataset and Models

3.1 Corpus

The analysis uses a corpus of 500 books spanning 20 genres from Project Gutenberg, including Adventure, Romance, Detective and Mystery, Science Fiction, and History. Each book is segmented into chunks of approximately 1,536 tokens with overlapping boundaries. The average number of chunks per book is approximately 94, with variation across titles. The Romance genre contains 25 books and approximately 3,018 chunks per model size, forming the target corpus for the genre deep-dive (Section 6).

3.2 Model Families

Five model families are evaluated:

- **Pythia (EleutherAI) [1]:** 8 checkpoints spanning 70M to 12B parameters, trained on the deduplicated Pile dataset. A consistent autoregressive architecture across all sizes makes this family well-suited for controlled emergence studies.
- **Cerebras-GPT [4]:** 7 checkpoints spanning 111M to 13B parameters, also trained on the Pile. A GPT-style autoregressive model with different training hyperparameters from Pythia.
- **Qwen2.5 [11]:** 4 checkpoints spanning 0.5B to 7B parameters. A more recent open-weight family with strong general capabilities and a qualitatively different training regime.
- **OpenAI `text-embedding-3-small` / `text-embedding-3-large`:** purpose-built retrieval embedding models included as an upper-bound reference point.
- **Qwen3-Embedding (0.6B / 4B / 8B):** an open-weight contrastive embedding family included to characterize the behaviour of embedding-trained models against base language models.

3.3 Embeddings

Pre-computed chunk embeddings are stored as PKL files, one file per book per model checkpoint. Pythia embeddings are 512-dimensional; Cerebras-GPT ranges from 768 to 4,096 dimensions depending on checkpoint size; Qwen2.5 from 896 to 3,584 dimensions. All embeddings are L2-normalized prior to any similarity computation.

4 Methods

4.1 Within vs Across Books Analysis

4.1.1 Formulation

For each book b in the corpus and each model checkpoint, a pool of P chunks is sampled from b (the *within* set) and an equal-sized pool of P chunks is sampled uniformly from all other books (the *across* set). Mean pairwise cosine similarity is computed within each set:

$$\begin{aligned}
\text{mean_within} &= \text{average cosine similarity among same-book chunks} \\
\text{mean_across} &= \text{average cosine similarity between same-book and other-book chunks} \\
\text{gap} &= \text{mean_within} - \text{mean_across} \\
\text{frac}_{>} &= \text{proportion of chunks with mean within-similarity} > \text{across-similarity}
\end{aligned}$$

A gap near zero indicates that the model places same-book passages no closer to each other than to random other-book passages, meaning book identity is not encoded in embedding geometry. A positive gap indicates the model has learned representations that preserve something about the source text.

4.1.2 Robustness Check

The analysis is repeated at three pool sizes ($P = 50, 100, 200$) to verify that results are not sensitive to sampling. Results are stable across all three pool sizes, confirming the signal is not a sampling artifact. Results reported here use $P = 200$.

4.2 Within-Book Consistency Analysis

As a preliminary analysis, we computed consecutive chunk similarity (cosine similarity between chunk i and chunk $i + 1$) and the fraction of each chunk's top-K nearest neighbours — within the same book — that fall within ±2 positions in reading order. Interactive HTML visualizations were generated per model size, displaying nearest-neighbour passages, UMAP/PCA projections, and HDBSCAN cluster hierarchies with TF-IDF topic labels. This analysis was intended to assess whether embedding spaces preserve local narrative structure.

4.3 BERTopic Topic Discovery

4.3.1 Design Rationale

The BERTopic pipeline uses pre-computed chunk embeddings as direct input rather than re-embedding from raw text. This means the topic clusters reflect the semantic similarity judgments of each model. Since the chunk text is identical across all model sizes, any difference in topic structure across checkpoints is attributable solely to the embedding representations.

4.3.2 Pipeline Configuration

Chunk embeddings are L2-normalized and passed into UMAP (`n_components=5`, `n_neighbors=15`, `metric=cosine`, `random_state=42`) for dimensionality reduction, then into HDBSCAN (`min_cluster_size=10`, `min_samples=5`, `metric=euclidean`, `cluster_selection_method=eom`) for density-based clustering. Topic representations are generated using c-TF-IDF applied to cluster-aggregated chunk texts, with a custom extended stopword list to suppress common narrative words (e.g., "said", "looked", "came") that would otherwise dominate topic labels.

4.3.3 Evaluation Metrics

The following metrics are tracked per model size:

- **Noise fraction:** percentage of chunks assigned to Topic -1. Lower values indicate the embedding space has sufficient density structure for HDBSCAN to operate.
- **Number of topics:** the total count of distinct topic clusters discovered.
- **Topics per book:** the average number of distinct topics spanned by a given book's chunks. Lower values indicate more concentrated, less fragmented clustering.
- **Books per topic:** the average number of distinct books contributing chunks to each topic. Higher values indicate genre-level themes rather than book-specific clusters.

4.4 Romance Genre Deep-Dive Methods

The genre deep-dive subsets the corpus to the 25 Romance books (∼3,018 chunks per checkpoint) and asks whether the corpus-level threshold also appears within a single genre. Five independent analyses are performed.

4.4.1 Within-vs-Across Genre Compactness

Analogous to the within-vs-across *books* analysis, but using genre as the partition. For each Romance chunk we compute mean cosine similarity to other Romance chunks (*within-genre*) and to a uniformly sampled set of non-Romance chunks (*across-genre*). We report the genre gap and a *cohesion ratio* (mean within-genre similarity divided by mean across-genre similarity) and a *separation gap* (the absolute difference). These two quantities decouple the signal from absolute similarity scale.

4.4.2 Nearest-Neighbour Lift

For each Romance chunk we compute the top-10 nearest neighbours over the full 500-book corpus by cosine similarity, and measure the fraction whose label matches the query along three axes:

- `lift10_genre`: fraction of neighbours sharing the query's coarse genre, divided by the corpus-wide chance baseline ($\approx 6.5\%$).
- `lift10_subject`: fraction sharing the query's primary Library of Congress subject heading, divided by chance ($\approx 0.4\%$).
- `lift10_subj_overlap`: cross-book Subjects-overlap lift; fraction of neighbours from a *different* book whose LoC subject set overlaps the query's, divided by chance ($\approx 0.8\%$).

Lift > 1 means above random; lifts are reported on the natural scale. Figure 1 shows the lift curves on the two richer label axes, and Figure 2 shows which genres Pythia treats as nearest to Romance.

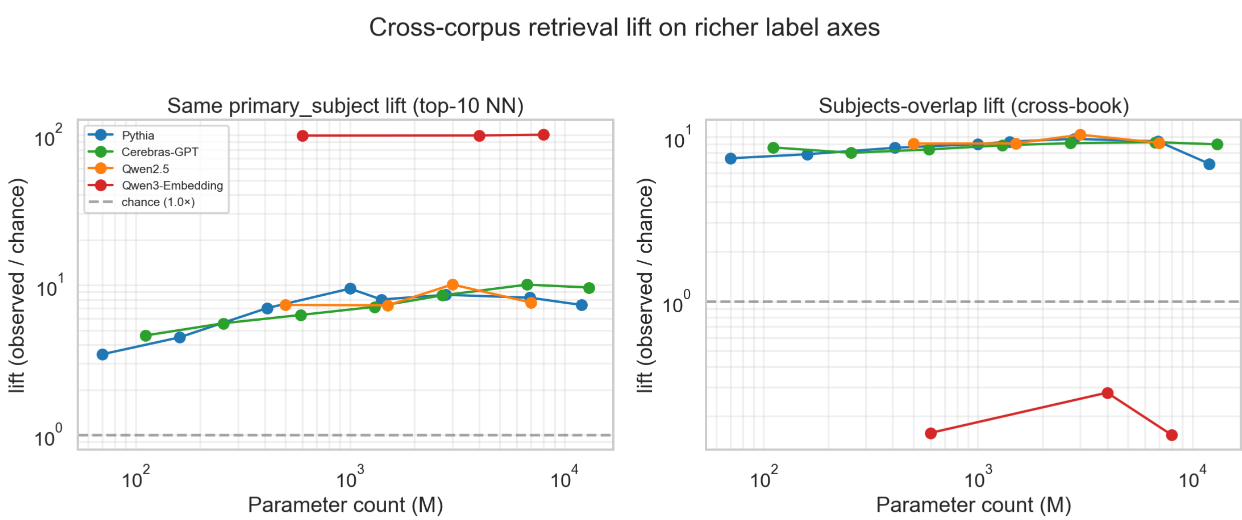

Figure 1: Cross-corpus retrieval lift on richer label axes. Same-subject lift grows strongly with scale for the base LMs; Qwen3-Embedding's cross-book Subjects-overlap lift sits *below* chance, exposing its within-document retrieval bias.

4.4.3 LLM-Labelled Topic Content

To test whether c-TF-IDF labels (which favour distinctive proper nouns such as character names) are obscuring the actual semantic content of clusters, we re-label every BERTopic topic using a local Mistral 7B model served via Ollama. For each topic we sample 5 representative chunks and prompt

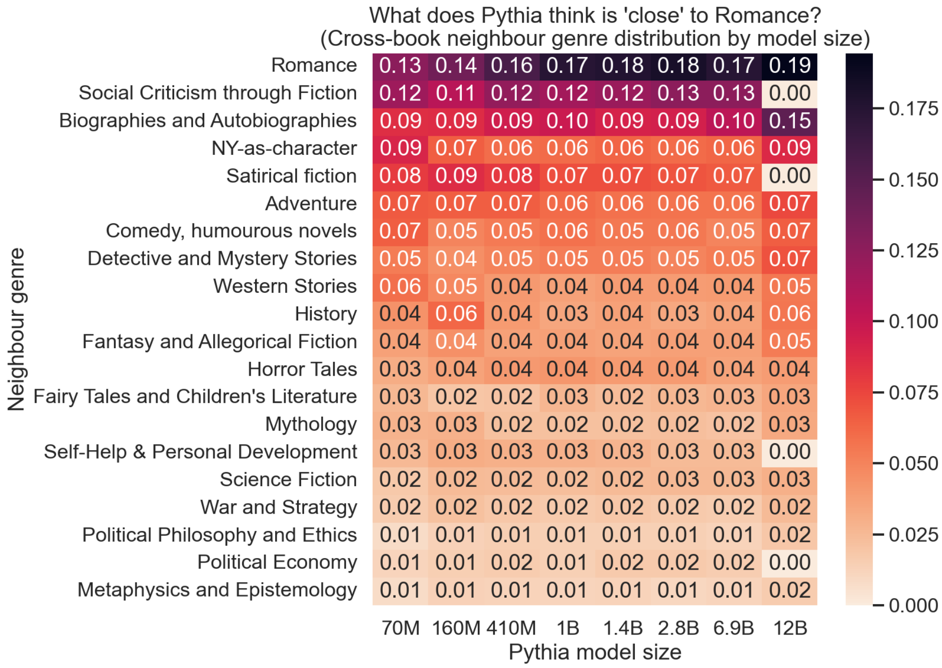

Figure 2: Cross-book nearest-neighbour genre distribution for Romance across Pythia sizes. Romance-to-Romance share grows (0.13 → 0.19); Social Criticism is consistently the second-ranked neighbour genre.

the model to produce a short thematic label and a list of theme words. We label 872 topics across 17 model checkpoints in roughly 15 minutes of local compute. We then compute three derived metrics:

- *Theme-word coverage*: fraction of LLM labels containing one of a curated list of Romance-relevant theme words (love, marriage, class, death, longing, etc.).
- *Multi-theme rate*: fraction of labels mentioning two or more disjoint themes (a hedge against unclean clusters).
- *Theme specificity*: a graded score combining the above two.

4.4.4 Categorical Alignment with Bookshelf and Book Labels

To distinguish geometric reorganization from label-aligned reorganization, we compute Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) between the BERTopic cluster assignment and (i) the Project Gutenberg bookshelf label and (ii) the source-book identifier. ARI and NMI both penalize granularity mismatch in different ways; reporting both makes the analysis robust to the (well-known) artifact that ∼50 topics versus ∼6 shelves drives ARI toward zero even under strong mutual information.

4.4.5 Reference Models

OpenAI `text-embedding-3-large` and Qwen3-Embedding (0.6B / 4B / 8B) are run through the same Romance deep-dive pipeline as upper-bound and contrastive reference points respectively.

5 Corpus-Level Results

5.1 Within-Book Consistency

The consecutive similarity metric yields high values at all Pythia sizes, ranging from 0.991 at 70M to 0.424 at 6.9B. However, this pattern does not reflect emergent narrative structure. At 70M and 160M, values above 0.97 are consistent with embedding collapse — all chunk vectors are nearly identical, making cosine similarity trivially high. The fraction of neighbours within ± 2 reading positions is flat at approximately 0.118 across all model sizes, indicating that top-K neighbourhood structure does not reflect reading-order proximity at any scale. This metric showed no sensitivity to model scale and was not pursued as a primary emergence signal. The outcome motivated the shift to the within-vs-across framework, which asks a structurally different question about book identity rather than sequential locality.

5.2 Within vs Across Books

Table 1 presents the within-vs-across results for all Pythia checkpoints.

Table 1: Pythia within-vs-across results (pool size $P = 200$).

Size	mean_within	mean_across	gap	$\text{frac}_>$
70M	0.996	0.996	0.0004	0.866
160M	0.984	0.983	0.0012	0.879
410M	0.737	0.703	0.0336	0.916
1B	0.621	0.582	0.0388	0.918
1.4B	0.488	0.451	0.0367	0.913
2.8B	0.482	0.437	0.0454	0.921
6.9B	0.418	0.366	0.0521	0.921
12B	0.416	0.372	0.0440	0.922

The results reveal a sharp threshold between 160M and 410M. The gap increases from 0.0012 to 0.0336 — a 28-fold increase in a single scale step. At 70M and 160M, absolute cosine similarities for both within and across groups are near unity (> 0.98), consistent with anisotropic embedding collapse. At 410M, absolute similarities decrease substantially (within = 0.737, across = 0.703), indicating the model is now producing geometrically differentiated embeddings where same-book similarity meaningfully exceeds other-book similarity.

The fraction of chunks for which within-book similarity exceeds across-book similarity is consistently high throughout (0.866–0.922), indicating that the direction of the book-identity effect is present even at small sizes. What is absent at 70M and 160M is *magnitude*, not direction.

Table 2 presents comparative results across model families.

Cerebras-GPT exhibits a gap of 0.047 at 111M — larger than Pythia achieves at 1.4B. No threshold transition is observable within the measured range, suggesting the transition falls below 111M for this architecture. The gap increases gradually with scale up to 6.7B before declining slightly at 13B.

Qwen2.5 presents a different pattern: the gap ranges from 0.014 to 0.027 with no clear monotonic trend. Absolute within-book similarities exceed 0.97 at all sizes, compressing the gap regardless of whether genuine book-identity structure exists. OpenAI embeddings achieve the largest gap by a

Table 2: Within-vs-across gap and fraction by model family.

Family	**Size**	**gap**	**frac$_>$**
Cerebras-GPT	111M	0.047	0.92
Cerebras-GPT	1.3B	0.058	0.94
Cerebras-GPT	6.7B	0.061	0.95
Cerebras-GPT	13B	0.052	0.94
Qwen2.5	0.5B	0.027	0.91
Qwen2.5	1.5B	0.014	0.88
Qwen2.5	3B	0.022	0.90
Qwen2.5	7B	0.020	0.90
OpenAI `te3-small`	—	0.151	0.998
OpenAI `te3-large`	—	0.174	0.999

substantial margin (0.174 for `text-embedding-3-large`), with frac$_>$ > 0.999, consistent with their explicit retrieval training objective.

5.3 BERTopic Topic Discovery on the Full Corpus

Table 3 presents BERTopic results across Pythia model sizes when restricted to the Romance genre (25 books, 3,018 chunks). We use Romance as the BERTopic target throughout because (i) it has enough density for meaningful clustering, (ii) it has known semantic overlap with adjacent genres (Social Criticism, Comedy), and (iii) it sets up the deep-dive in Section 6.

Table 3: BERTopic results by Pythia model size (Romance genre, 25 books, 3,018 chunks).

Size	**Noise frac.**	**# Topics**	**Topics/book**	**Books/topic**
70M	30.6%	∼50	∼31	∼15
160M	32.7%	∼50	∼31	∼15
410M	27.6%	∼45	∼28	∼16
1B	25.8%	∼45	∼27	∼17
2.8B	25.3%	∼45	∼26	∼17

Noise fraction at 70M and 160M (30.6% and 32.7%) is substantially higher than at 410M (27.6%) and subsequent sizes. *This transition mirrors the within-vs-across threshold precisely.* The two analyses share no assumptions, metrics, or intermediate computations; their agreement at the same scale boundary is therefore independent evidence of a genuine representational transition.

Topics per book decreases from approximately 31 at the small sizes to approximately 26 at 2.8B, indicating that a book's chunks become more concentrated across shared topic clusters rather than scattered into isolated, book-specific groups.

Qualitatively, topic keywords shift with scale. At 70M, the largest clusters mix character names from multiple different books — for example, one cluster contains characters from *Emma*, *Anna Karenina*, and *Pride and Prejudice* simultaneously, indicating no coherent thematic or book-level structure. At 410M, clusters become more book-coherent: one cluster is dominated by *Anna Karenina* characters (e.g., "alexey alexandrovitch"), another by Austen characters (e.g., "bingley · bennet · emma · knightley"). The c-TF-IDF labels take the form of character names because these are the terms most statistically distinctive to each cluster; the meaningful change is in the underlying geometric grouping, and Section 6.5 shows that the underlying themes are present at every scale once a non-frequency-based labeller is used.

Table 4 presents BERTopic results for Cerebras-GPT and Qwen2.5.

Table 4: BERTopic noise fraction and topics per book by family.

Family	**Size**	**Noise frac.**	**Topics/book**
Cerebras-GPT	111M	23.5%	41.0
Cerebras-GPT	1.3B	22.1%	28.5
Cerebras-GPT	6.7B	21.3%	26.8
Cerebras-GPT	13B	22.8%	27.4
Qwen2.5	0.5B	35.2%	33.1
Qwen2.5	1.5B	27.8%	29.0
Qwen2.5	3B	24.5%	27.8
Qwen2.5	7B	22.7%	26.5

Cerebras-GPT achieves low noise from its smallest checkpoint (23.5% at 111M), consistent with the within-vs-across finding that this family is already discriminative at 111M. Topics per book at 111M (41.0) is notably high relative to larger checkpoints, suggesting that while clustering is possible, the resulting topics are more fragmented at small sizes. Qwen2.5 starts with 35.2% noise at 0.5B and reaches 22.7% at 7B, a monotonic improvement that contrasts with the irregular pattern seen in its within-vs-across gap.

6 Romance Genre Deep-Dive

The corpus-level analyses establish a Pythia threshold near 410M. We now ask whether the same threshold appears at the chunk level inside a single genre, and what its character is. Five independent analyses are performed on Romance. Figure 3 gives a bird's-eye view of how the principal deep-dive metrics evolve with model size across all families.

6.1 Within-vs-Across Genre Compactness

Table 5 reports the within-genre vs across-genre gap for Pythia. The within-genre gap exhibits the same threshold as the within-book gap, jumping from 0.0007 at 160M to 0.024 at 410M — a 35-fold increase in one scale step. Cohesion ratio (mean within-genre similarity / mean across-genre similarity) follows the same shape: 1.01 at 160M $\rightarrow$ 1.26 at 410M, a 6.7$\times$ multiplicative jump on a quantity whose floor is 1.0. The conclusion is that Romance becomes a recognizable cluster in embedding space at the same scale at which book identity becomes recognizable.

6.2 Nearest-Neighbour Lift

Table 6 reports nearest-neighbour lift on three label axes. All three lifts are above chance at every scale, but they grow sharply through the 160M$\rightarrow$410M transition: `lift10_subject` jumps from 4.47$\times$ to 7.01$\times$ (chance = 0.4%), and the cross-book Subjects-overlap lift grows from 7.4$\times$ to 9.4$\times$. Concurrently, the mean top-1 cosine similarity drops from 0.992 at 160M to 0.843 at 410M — an 11-fold drop in $(1 - \text{mean\_cos\_top1})$, the cleanest single signature of the resolution of anisotropy.

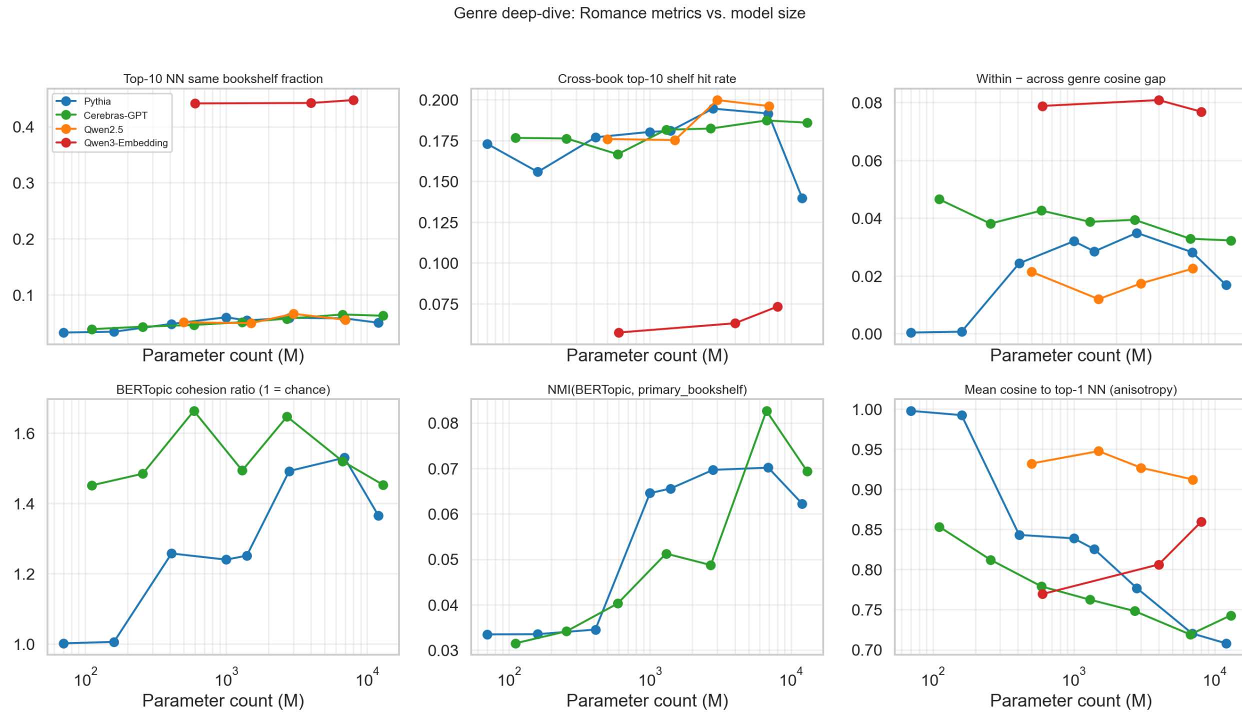

Figure 3: Romance deep-dive metrics versus model size across families. The Pile-trained families (Pythia, Cerebras-GPT) move sharply through the sub-1B region while Qwen2.5 and Qwen3-Embedding occupy distinct regimes.

6.3 Two Sequential Phases of Pythia Emergence

Combining the geometric metrics (gaps, ratios, anisotropy) with the categorical metrics (NMI/ARI against bookshelf and book labels) reveals that Pythia emergence does not happen in a single step. The transitions are sequential.

The interpretation is straightforward: at 160M→410M, anisotropy resolves and the embedding space acquires geometric room for genre-level structure to exist. At 410M→1B, the structure that exists begins to align with externally meaningful labels (bookshelves and books). Geometry comes first; label-aligned categorical structure follows. Figure 4 shows the two phases side by side.

6.4 Convergence: Six Independent Signals on Pythia 410M

Table 8 summarises the six independent signals that flag the same Pythia transition zone. Each row uses a different metric family — pairwise cosine gap, density-based clustering, retrieval lift, intra-cluster compactness, and absolute similarity scale — and yet they all change discontinuously between 160M and 410M.

6.5 LLM-Labelled Topic Content

A natural concern with the BERTopic results is that c-TF-IDF labels are dominated by character names — they pick the words most distinctive to a cluster, which in this corpus are proper nouns. This makes the labels look book-specific even when the underlying chunks share themes. To test this, we relabel every BERTopic topic using Mistral 7B served locally via Ollama, prompted on five

Table 5: Pythia within-vs-across *genre* compactness on Romance.

Size	Genre gap	Cohesion ratio	Separation gap
70M	0.0004	1.00	0.0007
160M	0.0007	1.01	0.0007
410M	0.024	1.26	0.024
1B	0.027	1.34	0.027
2.8B	0.031	1.44	0.031
6.9B	0.035	1.53	0.035

Table 6: Romance nearest-neighbour lift over the full 500-book corpus (Pythia). Chance baselines: genre ≈ 6.5%, subject ≈ 0.4%, subj-overlap ≈ 0.8%.

Size	lift10_genre	lift10_subject	lift10_subj_overlap	mean_cos_top1
70M	2.25	3.45	7.4	0.997
160M	2.41	4.47	7.6	0.992
410M	2.92	7.01	8.6	0.843
1B	3.10	7.65	8.9	0.776
2.8B	3.34	8.04	9.2	0.736
6.9B	3.55	8.23	9.4	0.708

representative chunks per topic. We label 872 topics across 17 model checkpoints in approximately 15 minutes.

The qualitative result: *the same kinds of themes appear at every scale*. Pythia 70M top topics include *Longing for a Better Life*, *Unrequited Love Conflicts*, *Death and Farewell*, *Social Class Transgression*. Pythia 6.9B top topics include *Struggle for Social Standing*, *Unrequited Love / Social Class Conflict*, *Marriage Proposals*. 88% of Pythia 70M LLM labels contain a core theme word.

What *does* scale is single-theme purity. Multi-theme hedging in the LLM labels shrinks for Pythia from 26% at 70M to 8% at 12B. For Cerebras the trend is reversed (21% → 40%), suggesting that Cerebras topics become more thematically mixed at large sizes despite the cleaner clustering geometry. Theme-word coverage stays high (70–88%) throughout for both Pile-trained families.

The conclusion is methodological as well as substantive: c-TF-IDF on a corpus of named characters *hides* the thematic content that the embeddings are actually clustering on. An LLM labeller given the chunks themselves recovers this content cleanly. Figure 5 shows the top LLM-labelled topics at four Pythia sizes; Figure 6 shows that c-TF-IDF theme specificity is flat across base-LM scale while OpenAI sits ∼3× higher; Figure 7 shows the label-quality signals across sizes.

6.6 Reference Models: OpenAI and Qwen3-Embedding

OpenAI `text-embedding-3-large` achieves ARI(BERTopic, bookshelf) = 0.17 versus the best Pythia value of 0.007 — a 25-fold improvement — and NMI(BERTopic, bookshelf) = 0.59 versus Pythia 0.07. Theme specificity reaches 0.36 versus 0.12 for the base LMs. These are the numbers expected of a model trained explicitly for retrieval. One caveat: the OpenAI BERTopic run used ∼21 chunks/book versus ∼120 for the Pythia/Cerebras runs, which produces a one-topic-per-book artifact in some clusters; the ARI/NMI gains are nevertheless robust to this.

Table 7: Geometric vs categorical phase transitions in Pythia Romance.

Phase	Signal	Pre	Post	Multiplier
Geometric (160M→410M)	Cohesion ratio	1.01	1.26	6.7×
	Separation gap	0.0007	0.024	3.5×
	1 − mean_cos_top1	0.008	0.157	11.0×
Categorical (410M→1B)	NMI(BERTopic, bookshelf)	0.034	0.064	29.3×
	NMI(BERTopic, book)	0.116	0.173	17.6×

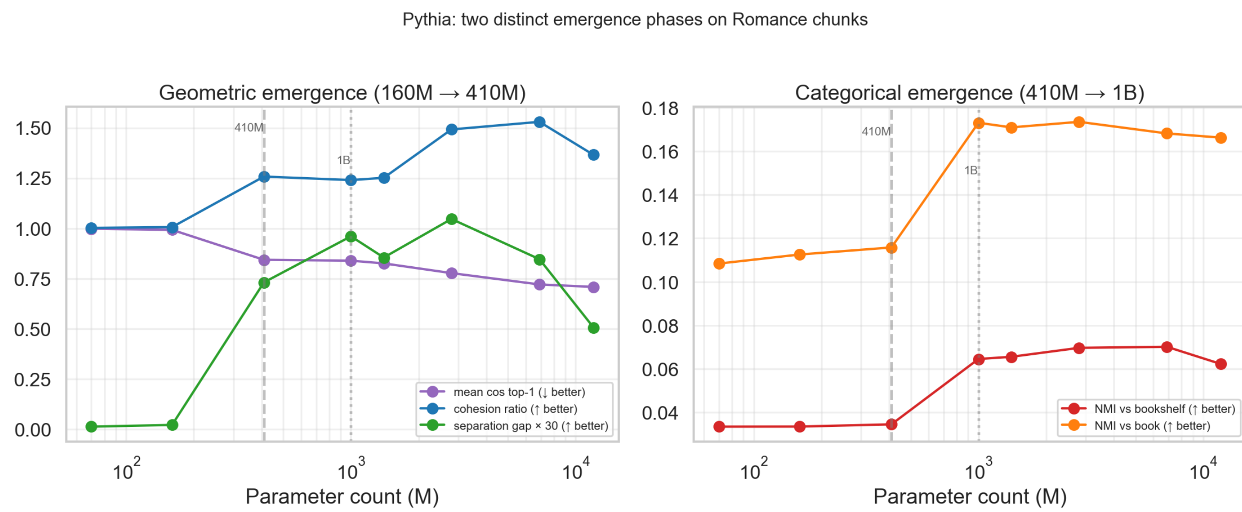

Figure 4: Two distinct emergence phases in Pythia on Romance chunks. Left: geometric emergence (cohesion ratio, separation gap, anisotropy) resolves at 160M→410M. Right: categorical emergence (NMI against bookshelf and book labels) follows at 410M→1B.

Qwen3-Embedding (0.6B / 4B / 8B) occupies a qualitatively different regime. 95% of each chunk's top-10 nearest neighbours come from the *same book*, yielding a same-book concentration lift of ∼300× versus 7–25× for the base LMs. At the same time, its cross-book Subjects-overlap lift collapses to 0.15× — *below chance*. Interpreted operationally, Qwen3-Embedding has been trained to function as a within-document retriever: it is excellent at locating other passages *from the same book* but gives up cross-book semantic generalization to do so. This is a different optimization target from the OpenAI family and from the base LMs, and the within-genre framework cleanly exposes the trade-off. Figure 8 makes the contrast visually explicit.

7 Discussion

7.1 The 410M Threshold in Pythia

The convergence of six independent methods on the same Pythia scale boundary is the central finding of this study. The within-vs-across book gap increases 28-fold between 160M and 410M; the within-vs-across genre gap increases 35-fold; the BERTopic noise fraction drops; nearest-neighbour subject lift roughly doubles; cohesion ratio crosses 1.2; and the inverse-anisotropy quantity 1 − mean_cos_top1 jumps 11-fold. These analyses share no common assumptions: the within-vs-across pipelines operate on pairwise cosine similarities between raw chunk vectors without any clustering; the BERTopic

Table 8: Six independent metrics flag Pythia 410M as the emergence point.

Signal	**Pre (70M–160M)**	**Post (410M+)**
Within-vs-across BOOK gap	0.0004–0.0012	0.034–0.055
BERTopic noise fraction	30–33%	25–28%
Within-vs-across GENRE gap	0.0004–0.0007	0.024–0.035
`lift10_subject` (NN)	3.45–4.47×	7.01–9.47×
Cohesion ratio (BERTopic)	1.00–1.01	1.26–1.53
mean_cos_top1 ↓ (anisotropy)	0.992–0.997	0.708–0.843

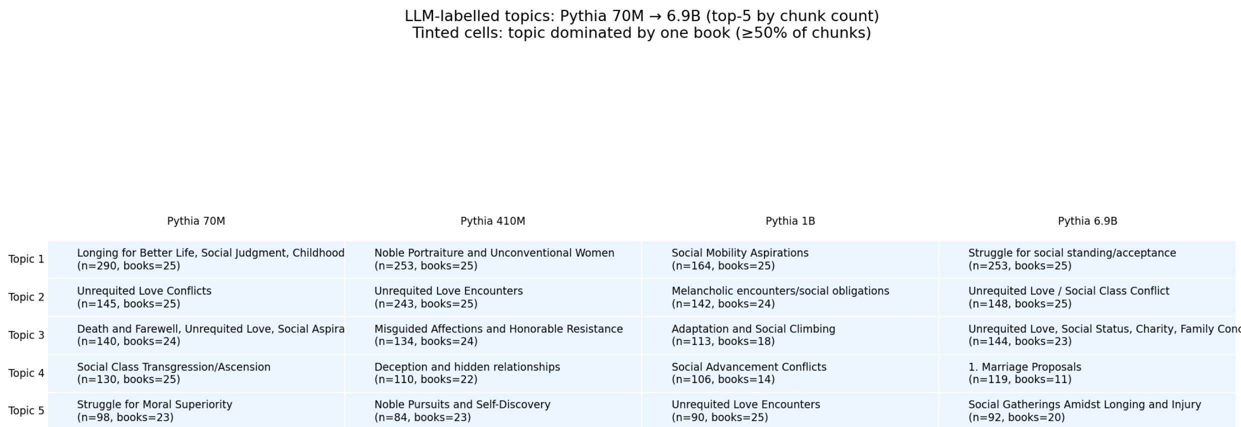

Figure 5: Top-5 LLM-labelled topics by chunk count for Pythia 70M, 410M, 1B, and 6.9B. The same kinds of themes (longing, unrequited love, social class, marriage) appear at every scale.

pipeline applies density-based clustering in a UMAP-reduced subspace; the nearest-neighbour analysis computes per-query lift over external label axes. Their agreement therefore constitutes strong, method-robust evidence that a genuine representational transition occurs at approximately 400M parameters in Pythia.

The geometric interpretation of this transition is consistent with the resolution of embedding anisotropy. At 70M and 160M, cosine similarities between any two chunks — regardless of source — are above 0.98. This is the signature of a severely anisotropic embedding space in which all vectors are nearly collinear. At 410M, absolute within-book similarities drop to 0.737, indicating that the model now uses a substantially larger fraction of its representational capacity. This geometric expansion is what allows the cosine gap, the HDBSCAN density structure, and the nearest-neighbour lift to all become informative simultaneously.

7.2 Two Sequential Phases of Emergence

The Romance deep-dive separates the threshold into two sequential events: a geometric phase in which anisotropy resolves and a categorical phase in which discovered structure begins to align with externally meaningful labels. This decoupling matters because it implies that label-

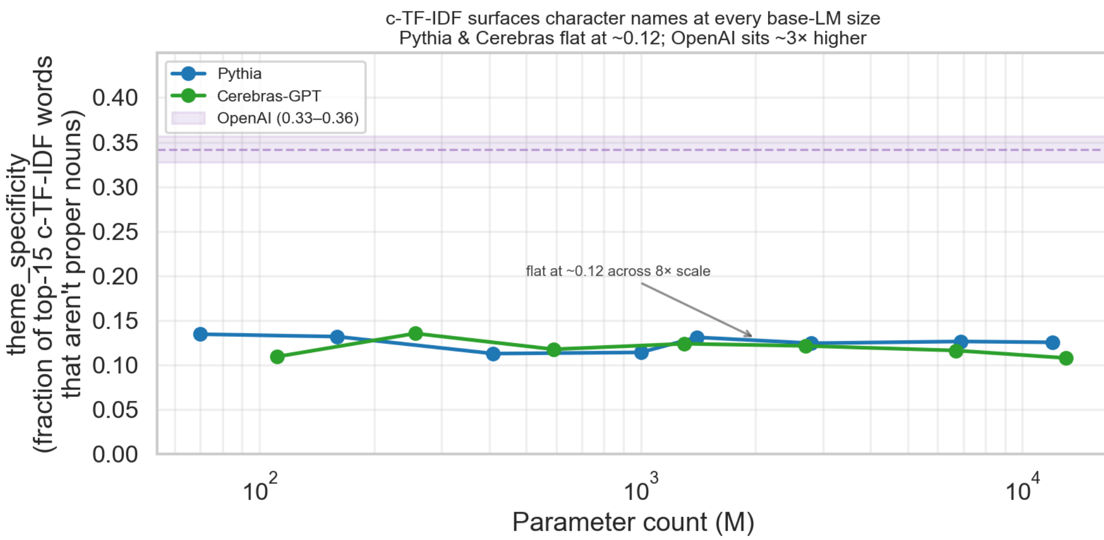

Figure 6: c-TF-IDF theme specificity is flat at ∼0.12 across an 8× scale range for both Pile-trained families, while OpenAI sits in the 0.33–0.36 band — evidence that c-TF-IDF, not the embeddings, suppresses thematic content.

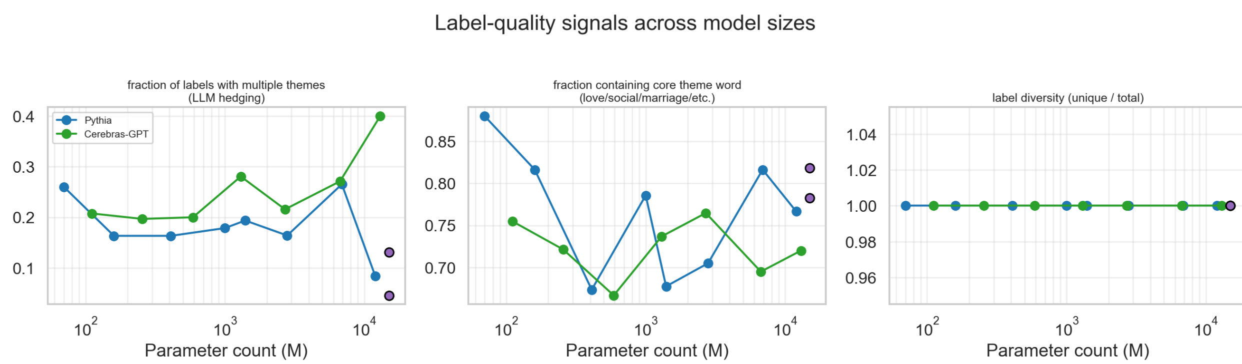

Figure 7: Label-quality signals across model sizes. Multi-theme hedging shrinks for Pythia (26% → 8%) but grows for Cerebras; core-theme-word coverage stays high (∼70–88%) throughout.

aligned emergence is downstream of, not coincident with, geometric emergence. A model can have geometrically meaningful embeddings before those embeddings agree with any human-curated taxonomy, and indeed Pythia at 410M is an existence proof.

7.3 Architecture-Dependent Thresholds

The contrast between Pythia and Cerebras-GPT is instructive. Both are autoregressive models trained on the same Pile dataset, yet Cerebras shows no threshold within the measured range, being already discriminative at 111M. The within-vs-across gap at Cerebras 111M (0.047) exceeds what Pythia achieves at 1.4B. This indicates that the scale threshold is not a universal function of parameter count but depends on architecture details or training hyperparameters. Identifying which specific factor is responsible would require controlled ablation experiments beyond the scope of this study.

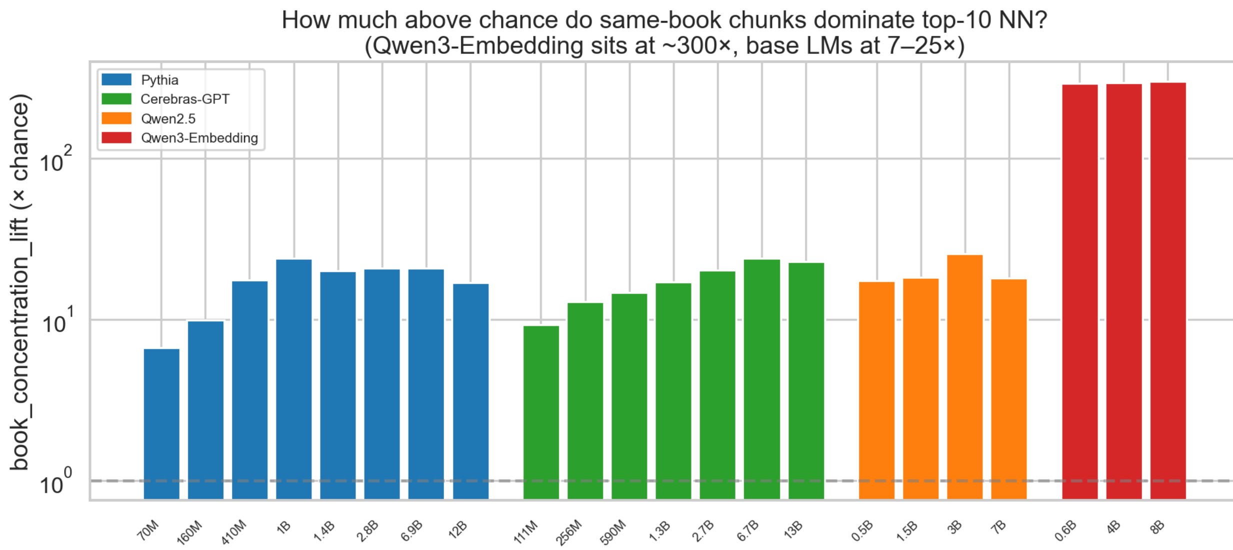

Figure 8: Same-book concentration lift (log scale). Base LMs sit at 7–25× chance; Qwen3-Embedding sits at ∼300×, behaving as a within-document retriever rather than a within-genre one.

7.4 Qwen2.5 and the Compressed-Similarity Regime

Qwen2.5's uniformly high absolute cosine similarities (within > 0.97, across > 0.94 across all sizes) represent a qualitatively different geometric regime from both Pile-trained families. The within-vs-across gap is compressed throughout, not because book identity is absent, but because the similarity scale itself is compressed — all pairs are pushed toward high similarity regardless of semantic relationship. The BERTopic noise fraction for Qwen2.5 does decrease monotonically from 35.2% to 22.7% across sizes, suggesting that clustering structure improves with scale even when the cosine gap metric cannot detect it. This is methodologically important: a single emergence metric can fail to register a genuine transition if it interacts pathologically with the underlying similarity geometry, which is precisely why the convergence framework matters.

7.5 Embedding Models Are a Different Animal

The contrast between OpenAI `text-embedding-3` and Qwen3-Embedding makes clear that purpose-trained embedding models are not simply "better base LMs." OpenAI's family behaves like a maximally well-organized version of the base-LM regime, with the same qualitative geometry but much sharper cluster structure. Qwen3-Embedding behaves *differently*: it is a within-book retriever whose cross-book semantic generalization is below chance. Both are reasonable optimization targets, but they are not the same target, and a single corpus-level summary statistic would conflate them. The within-genre framework introduced here cleanly distinguishes the two.

7.6 Limitations

Several limitations should be acknowledged. The Romance genre analysis involves 25 books and approximately 3,000 chunks, which limits statistical power for fine-grained comparisons. The BERTopic hyperparameters (UMAP `n_components=5`, HDBSCAN `min_cluster_size=10`) are held constant across all model sizes; a more rigorous analysis might separately optimize parameters per family. No formal statistical significance tests (e.g., permutation tests for the within-vs-across gap) were conducted. The consistency of results across three pool sizes, the convergence of six independent metrics, and the qualitative agreement of the LLM labels provide informal robustness evidence, but statistical quantification remains a gap.

8 Future Work

8.1 Statistical Validation

A natural extension is to compute permutation-based p-values for the within-vs-across gap by randomly reassigning chunk-to-book labels and recomputing the gap distribution. This would provide formal significance estimates and confidence intervals for the threshold location. Bootstrap sampling over books would additionally quantify how sensitive the threshold location is to the specific books included in the corpus.

8.2 Extension to Additional Model Families

The current analysis covers Pythia and Cerebras-GPT from the Pile training distribution and Qwen2.5 from a different regime. Extending the pipeline to additional families, particularly families with known architectural differences (e.g., encoder-only models, models with different attention variants) would help isolate which training or architectural factors determine the threshold location. Comparing models trained on different datasets at matched parameter counts would be particularly informative.

8.3 Genre-Level Analysis Across All Genres

The deep-dive was conducted on Romance only. Extending the five-angle pipeline to all 20 genres and comparing convergence patterns across genres would reveal whether the two-phase structure is genre-invariant or whether narrative richness, vocabulary diversity, and inter-genre overlap shift the threshold location. Genres with distinctive vocabulary (Horror, Science Fiction, War & Strategy) are likely to show the geometric phase earlier than genres with generic vocabulary (Social Criticism, Biographies).

8.4 Subword-Level Embedding Geometry

The current analysis treats chunk embeddings as atomic units. A complementary direction would be to examine token-level or subword-level embedding similarity distributions, connecting the macro-level emergence signal to lower-level representational changes already documented in the mechanistic interpretability literature.

8.5 Scaling the LLM-Labeller

The Mistral 7B labelling pipeline labels 872 topics in 15 minutes on a single machine. Scaling this to label every chunk (rather than every topic) would allow the construction of model-agnostic "ground-truth" theme labels and a direct evaluation of how well each model's clusters track them, decoupling cluster quality from labelling quality.

9 Conclusion

This study presents evidence that representational emergence in language model embeddings is real, quantitatively locatable, method-robust, and structured into two sequential phases — at least within the Pythia model family. Our key contributions are as follows:

1. A within-vs-across books analysis pipeline that identifies a sharp threshold between 160M and 410M Pythia parameters, where the ability to encode book identity at the passage level increases 28-fold in a single scale step.
2. A BERTopic topic discovery pipeline that operates directly on pre-computed chunk embeddings, producing independent corroboration of the same threshold through a drop in HDBSCAN noise fraction at the same model size.
3. A Romance genre deep-dive along five independent angles that locates the same threshold inside a single genre and decomposes it into a geometric phase (160M→410M) followed by a categorical phase (410M→1B).
4. Six metrics across four metric families (pairwise gap, density clustering, retrieval lift, anisotropy) that all flag Pythia 410M, providing the strongest method-robust evidence available without additional model training.
5. A demonstration via local LLM-based topic labelling that thematic content is present at every scale and that c-TF-IDF was hiding it; what scales is *single-theme purity*, not the existence of themes.
6. A comparative analysis demonstrating that Cerebras-GPT achieves discriminative capability below its smallest measured checkpoint, that Qwen2.5 occupies a compressed-similarity regime in which the cosine-gap metric is unreliable, and that purpose-trained embedding models are categorically different: OpenAI is a sharper version of the base-LM regime, while Qwen3-Embedding is a within-book retriever whose cross-book generalization is below chance.

From a methodological standpoint, the label-free nature of the embedding-side pipelines is a strength: neither requires genre metadata, author annotations, or external taxonomies during analysis. Labels are used only *post hoc* to score whether the unsupervised structure recovers them. The signal is derived purely from embedding geometry and is therefore applicable to any corpus and any model family without curated supervision. This positions the within-vs-across approach, in combination with the genre-level convergence framework, as a complement to label-dependent clustering benchmarks and a practical diagnostic for evaluating embedding quality in contexts where human labels are unavailable.

A Topic × Bookshelf Confusion

Figure 9 shows the BERTopic-topic × bookshelf confusion for Pythia 70M versus 6.9B. The granularity mismatch ($\sim$50 topics against $\sim$6 shelves) is why ARI stays near zero even as NMI rises with scale: many fine-grained topics map into a single shelf, which ARI penalizes but NMI does not.

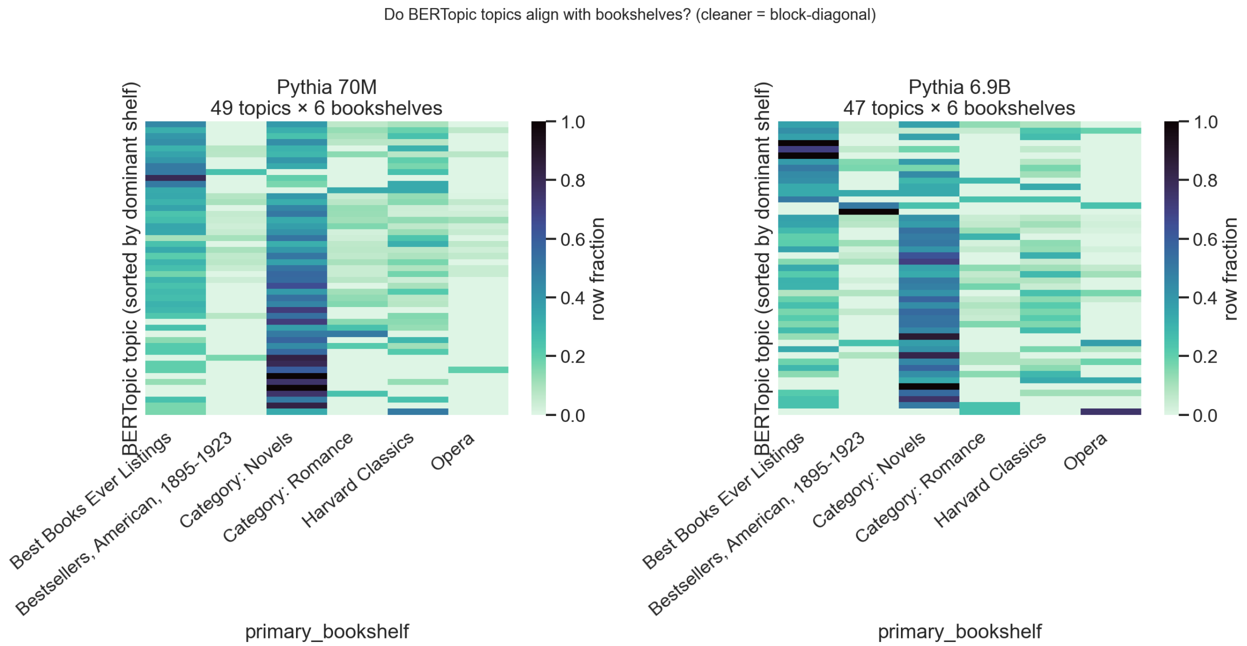

Figure 9: BERTopic topic × primary-bookshelf row-fraction heatmaps for Pythia 70M (left) and 6.9B (right). Larger models produce a more block-diagonal structure.

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